01. acf_force_control_simple

ACTIVE CONTACT FLANGE | Workpiece normal pressing force
Voice coil or leadscrew; native force reconstruction, PI/PID and measured-motion feedforward
Edit p in the Model Workspace. Use acf_run for metrics and HTML replay.



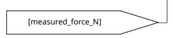
Actuator selection

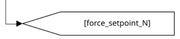
Force controller selection

FAST FORCE LOOP — measured force and S inertia, with no surface preview



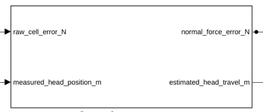
Requested force





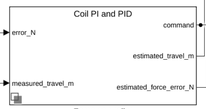


Force error

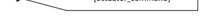


Contact force measurement





Force controller







Actuator and RABS mechanics



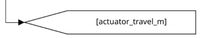




Force measurement









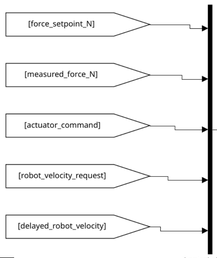




Relative position measurement



RECORDINGS AND WORKPIECE FORCE SCOPE



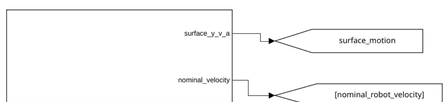
Control signals



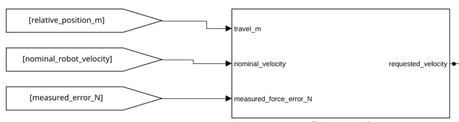


Control log

SLOW ROBOT LOOP — force must remain stable before a recentering command



Surface and nominal path

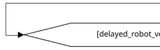


Recenter travel





Robot command delay





Robot motion





Plant log



Robot log



Surface log

CAUSAL ROBOT PREDICTION — commands already issued, with calibrated delay and response





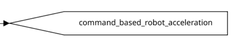
Issued-command robot predictor

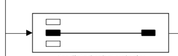

Predicted robot acceleration


Robot prediction sampling



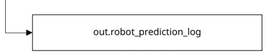
Enable robot acceleration prediction



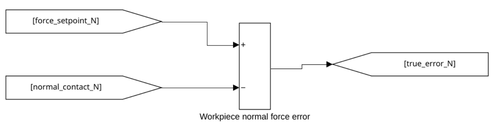

Predicted robot velocity


Predicted velocity sampling





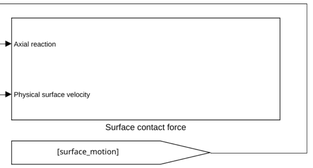
Command prediction diagnostics



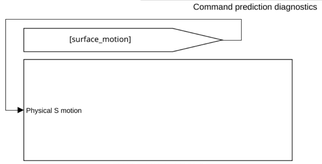
Workpiece normal force error

CONTACT PHYSICS AND SENSORS — actual motion enters the plant and sensor models

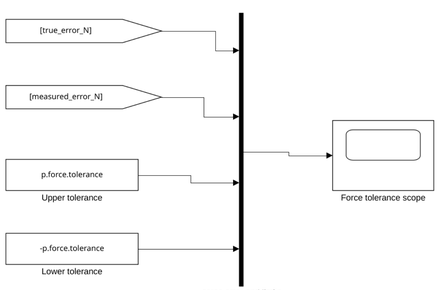




Surface contact force



S accelerometer and ADC



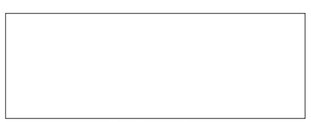
Upper tolerance
Lower tolerance

Force tolerance scope

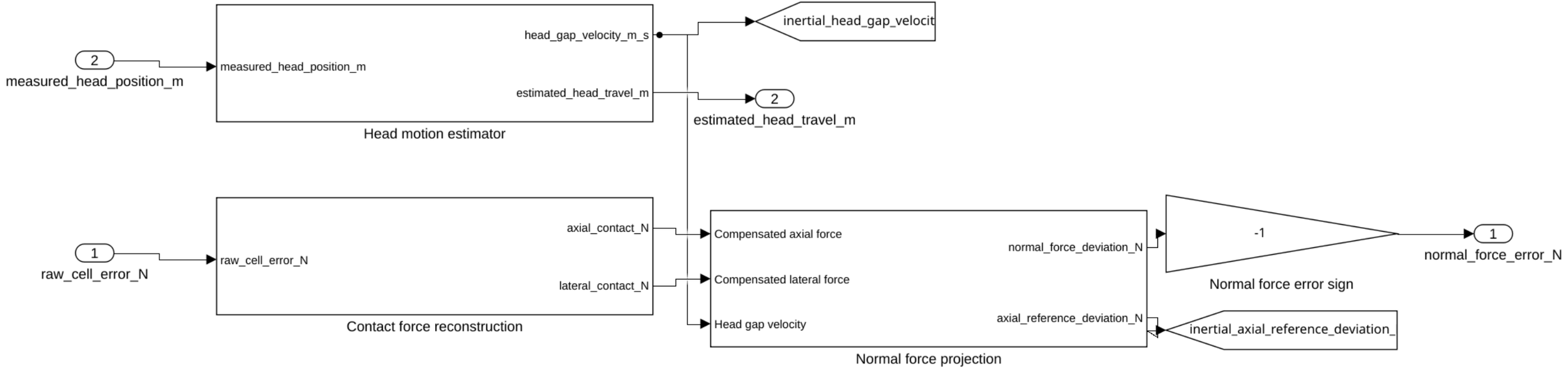
Force error and limits


Lateral B-S force measurement

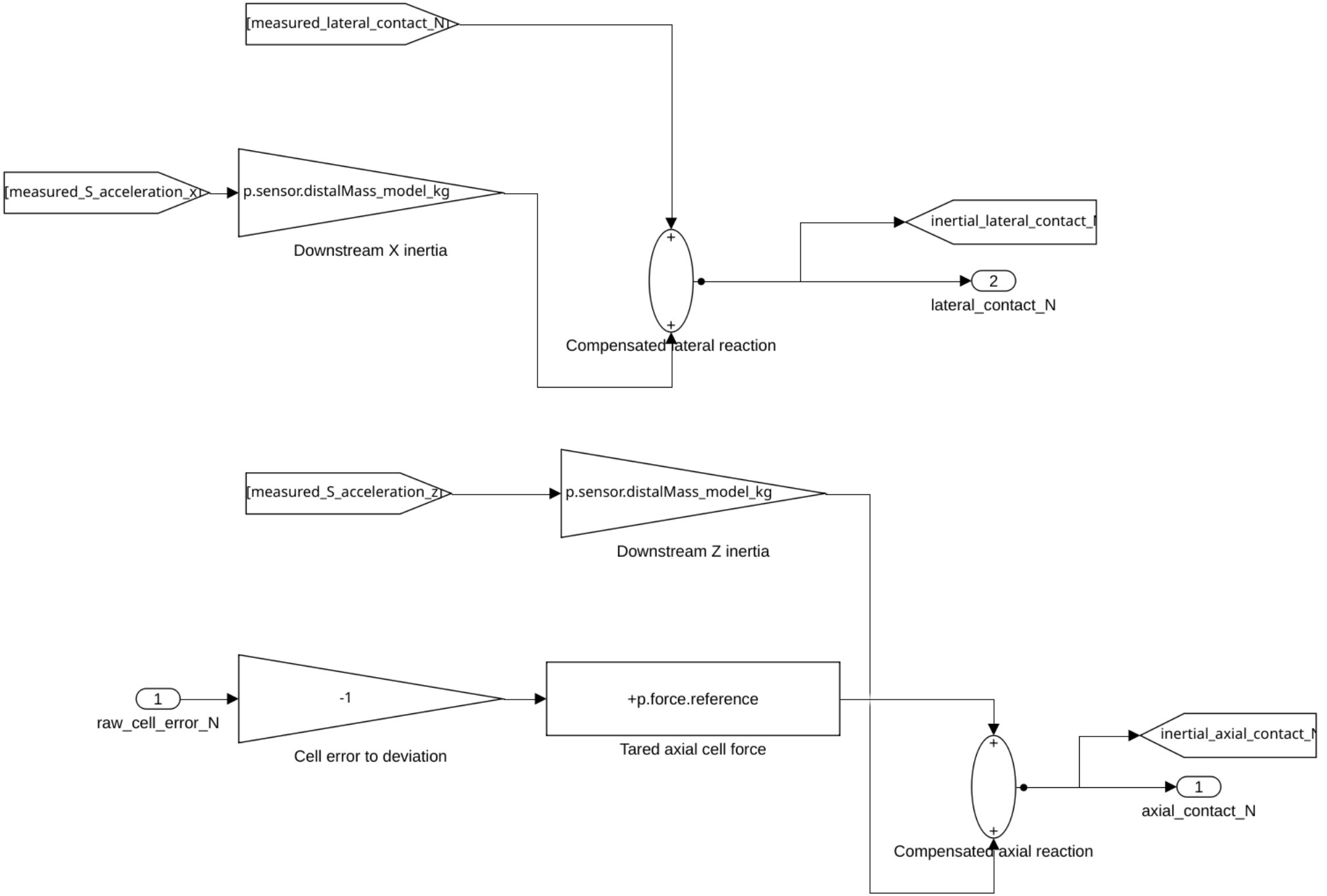

Inertial sensor recordings


Normal force recordings

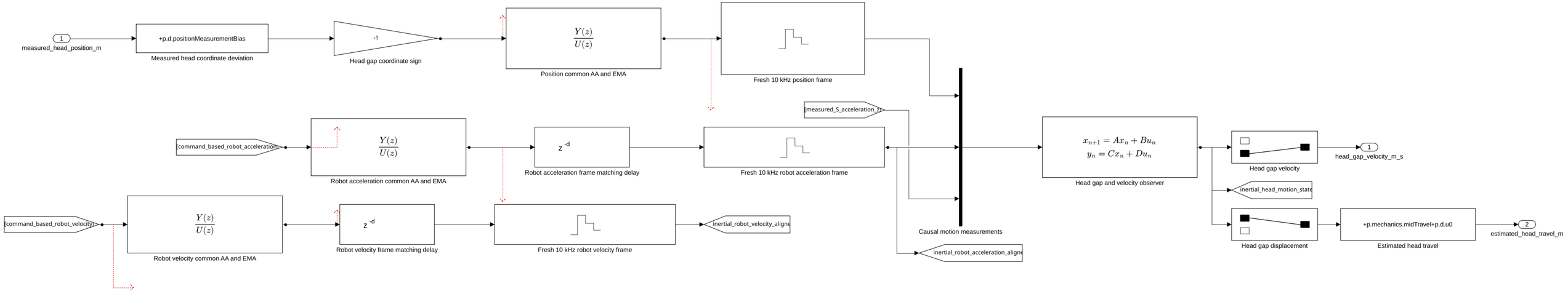
02. acf_force_control_simple/Contact force measurement

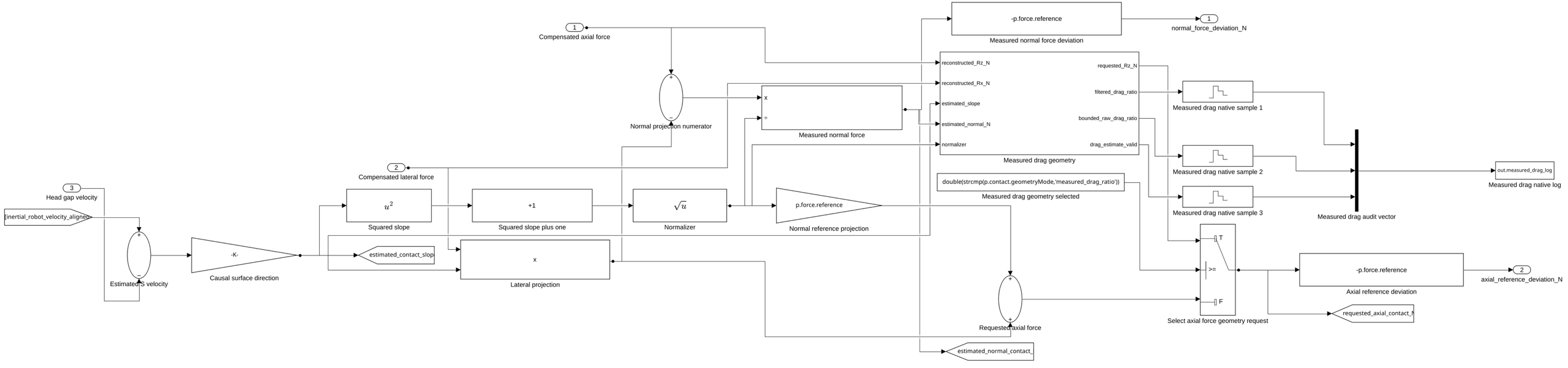


03. acf_force_control_simple/Contact force measurement/Contact force reconstruction

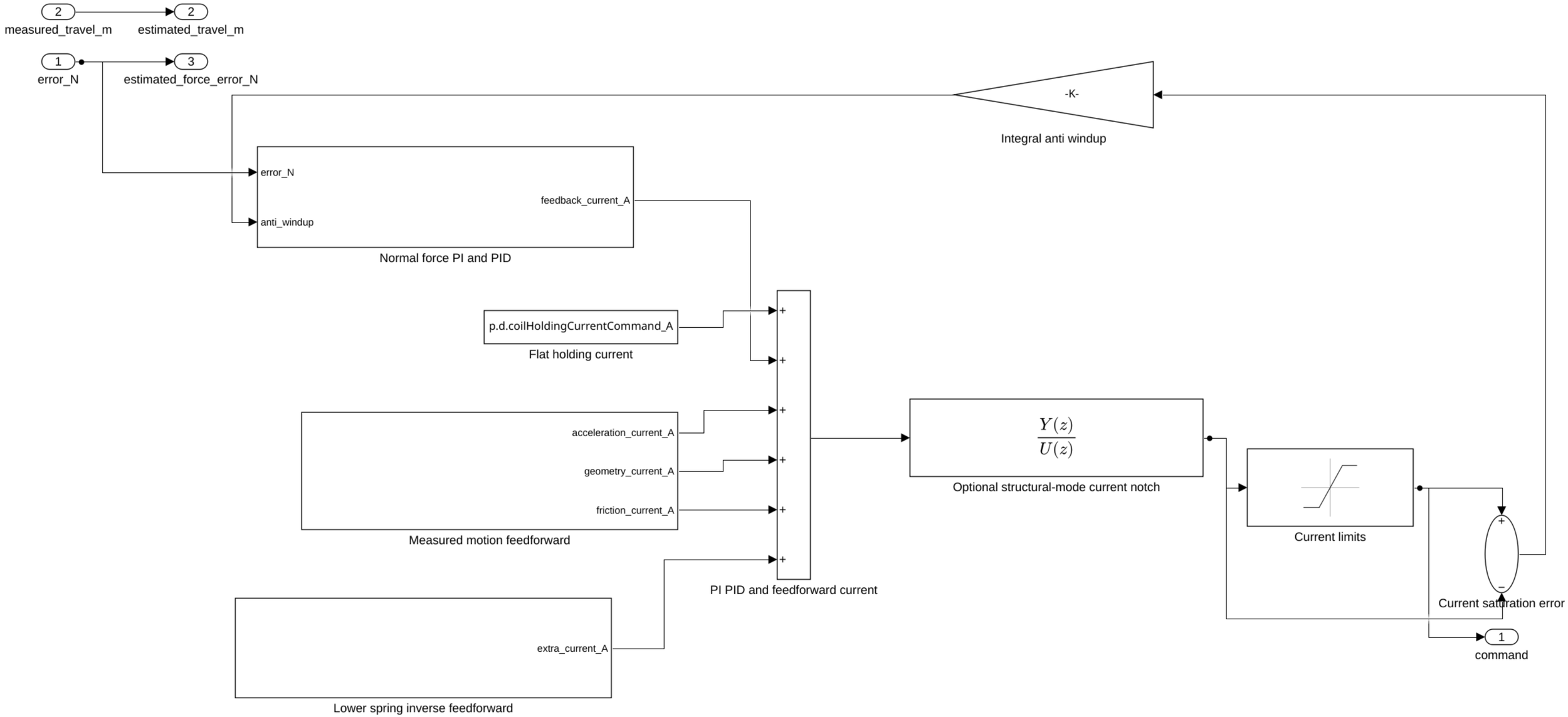


04. acf_force_control_simple/Contact force measurement/Head motion estimator

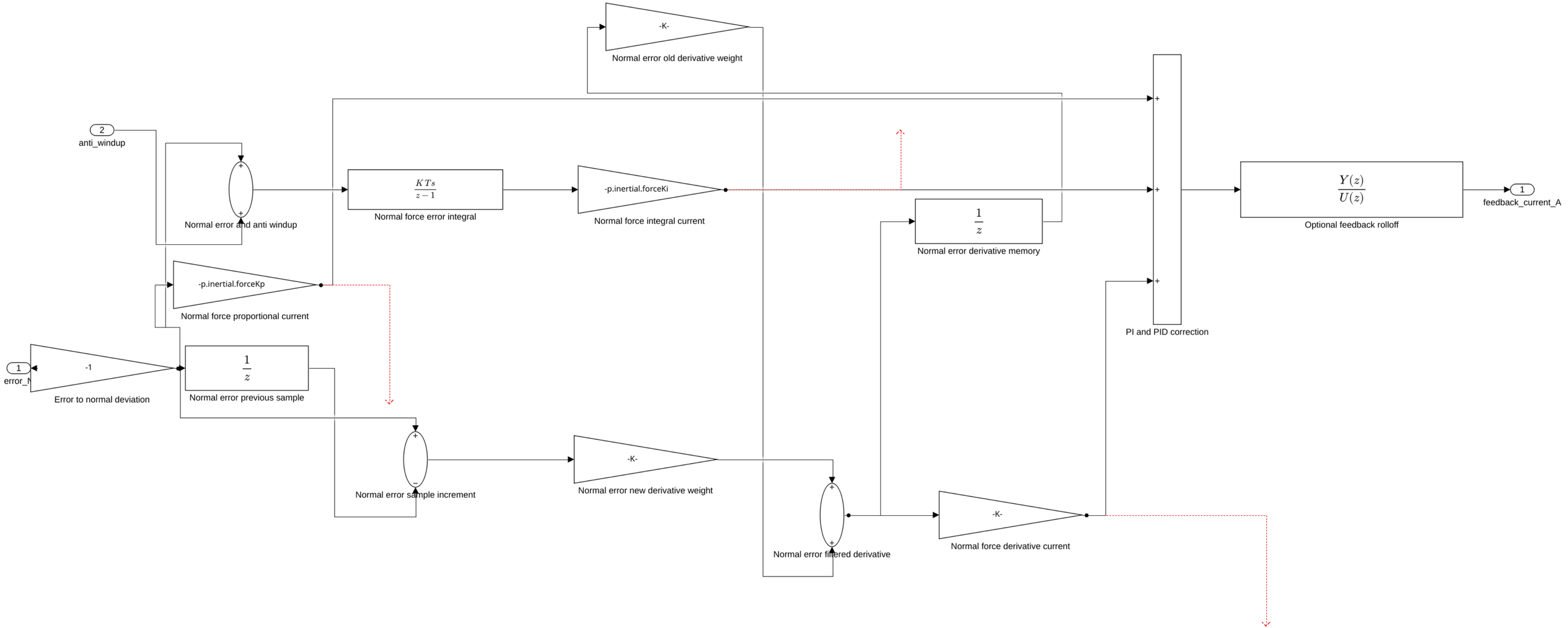



1
Compensated axial force
-p.force.reference
Measured normal force deviation
1
normal_force_deviation_N
reconstructed_Rz_N
reconstructed_Rx_N
estimated_slope
estimated_normal_N
normalizer
requested_Rz_N
filtered_drag_ratio
bounded_raw_drag_ratio
drag_estimate_valid
Measured drag geometry
Measured drag native sample 1
Measured drag native sample 2
Measured drag native sample 3
Measured drag audit vector
out.measured_drag_log
Measured drag native log
Normal projection numerator
x
÷
Measured normal force
2
Compensated lateral force
double(strcmp(p.contact.geometryMode,'measured_drag_ratio'))
Measured drag geometry selected
3
Head gap velocity
inertial_robot_velocity_aligned
Estimated TS velocity
-K-
Causal surface direction
u^2
Squared slope
+1
Squared slope plus one
$\sqrt{u}$
Normalizer
p.force.reference
Normal reference projection
estimated_contact_slop
x
Lateral projection
Requested axial force
[] T
>=
[] F
Select axial force geometry request
-p.force.reference
Axial reference deviation
2
axial_reference_deviation_N
requested_axial_contact_f
estimated_normal_contact_

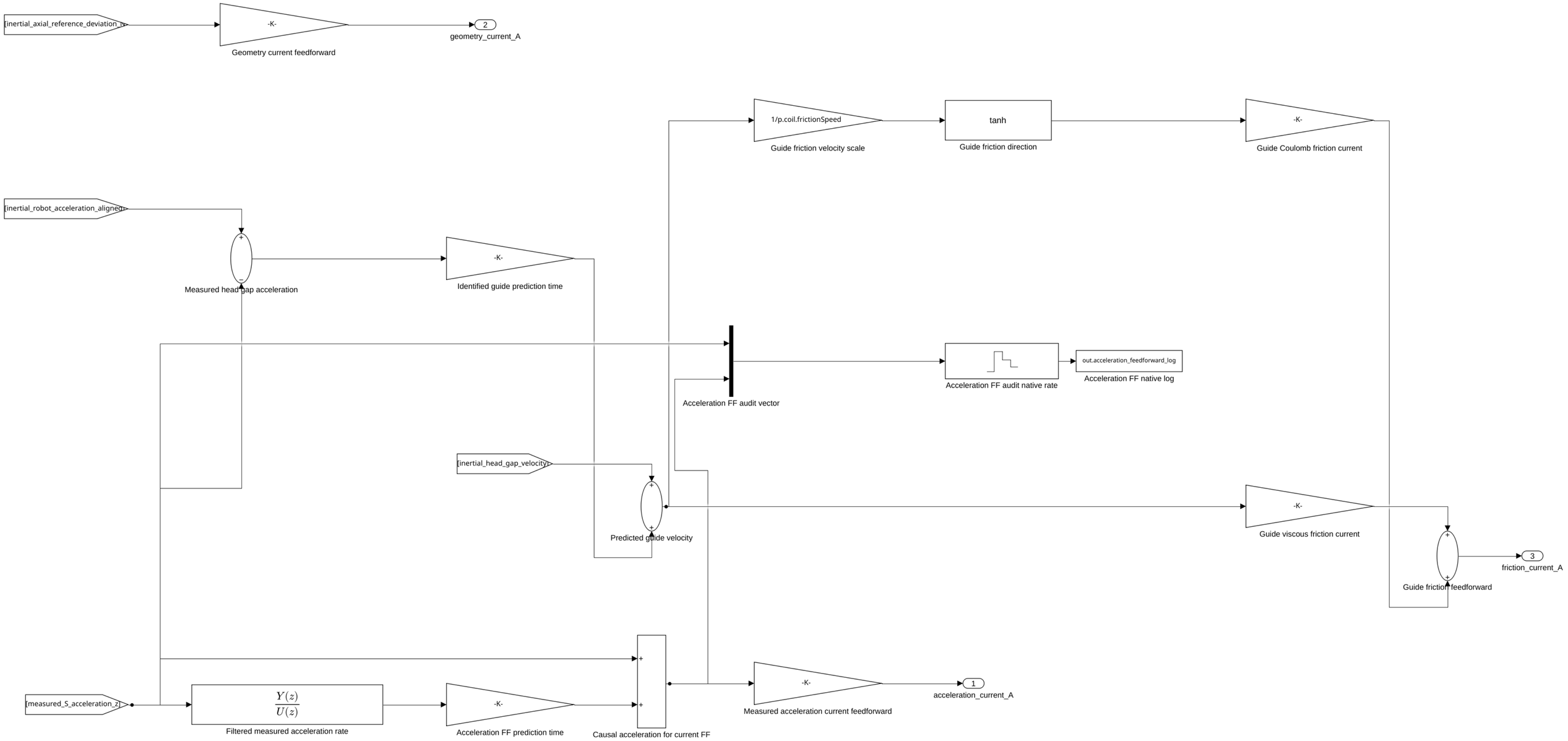
06. acf_force_control_simple/Force controller/Coil PI and PID



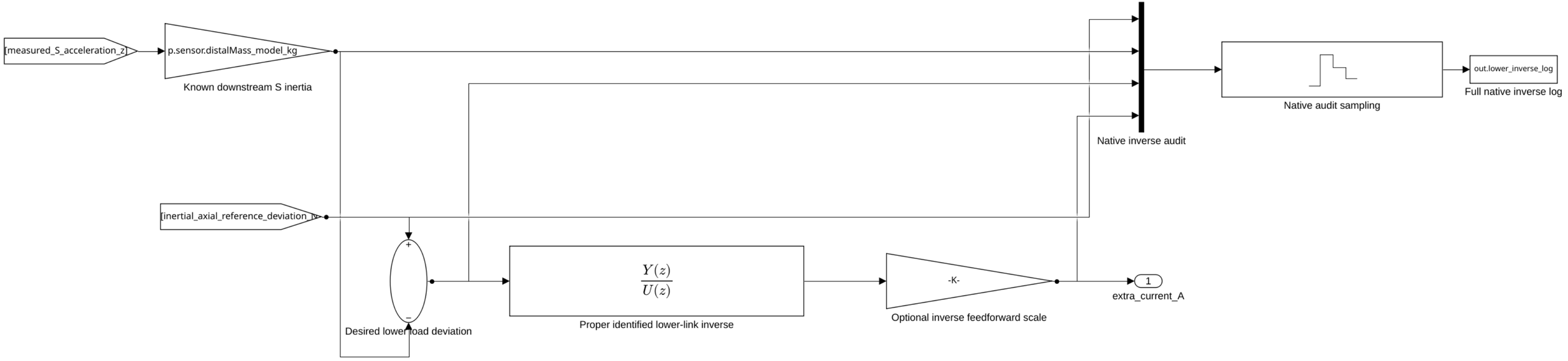
07. acf_force_control_simple/Force controller/Coil PI and PID/Normal force PI and PID



08. acf_force_control_simple/Force controller/Coil PI and PID/Measured motion feedforward

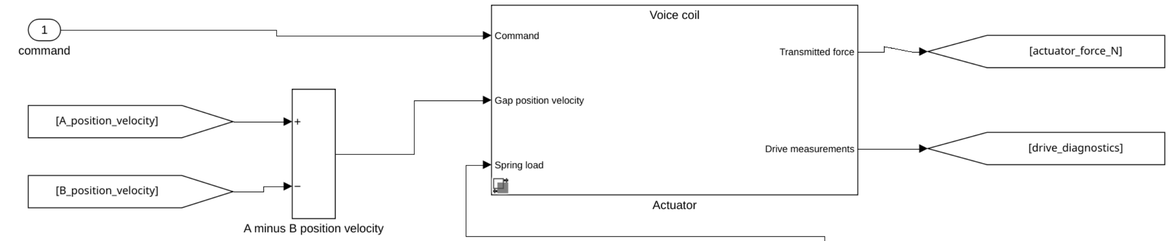


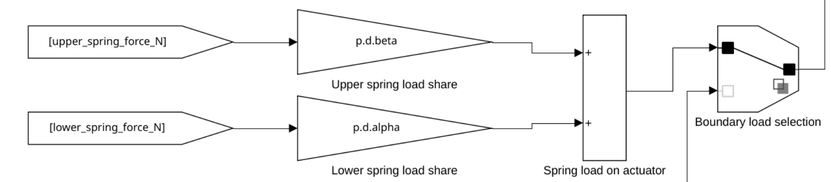
09. acf_force_control_simple/Force controller/Coil PI and PID/Lower spring inverse feedforward

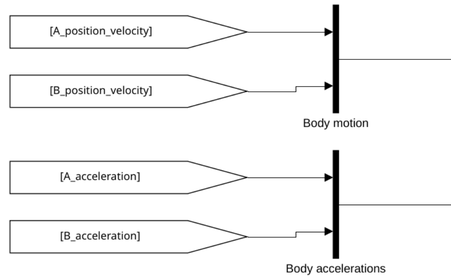


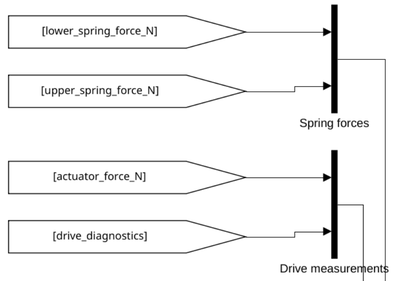
10. acf_force_control_simple/Actuator and RABS mechanics

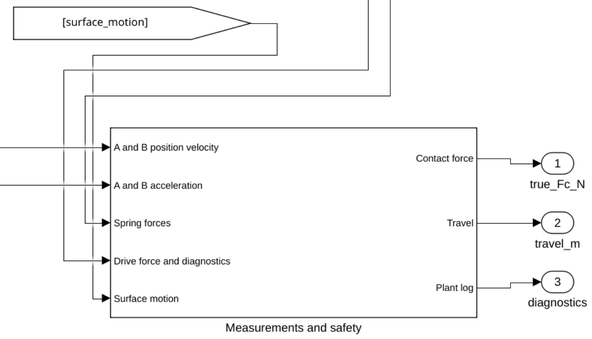
ACTUATOR AND MECHANICS — native actuator, A/B masses and optional links

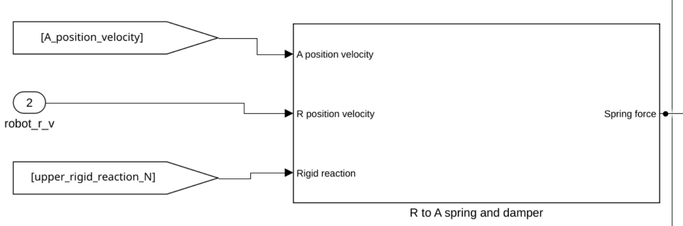




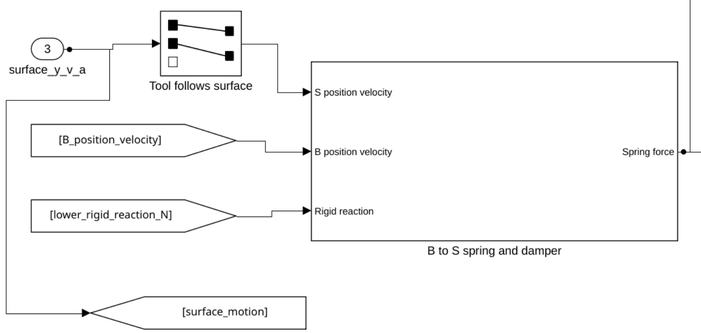


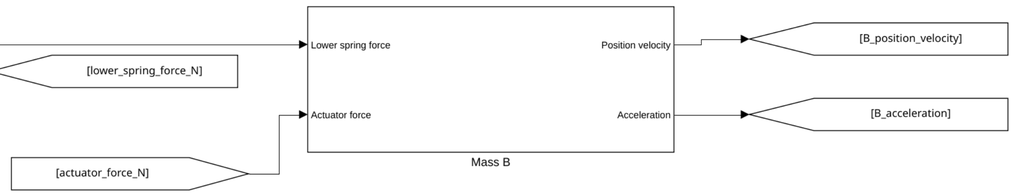


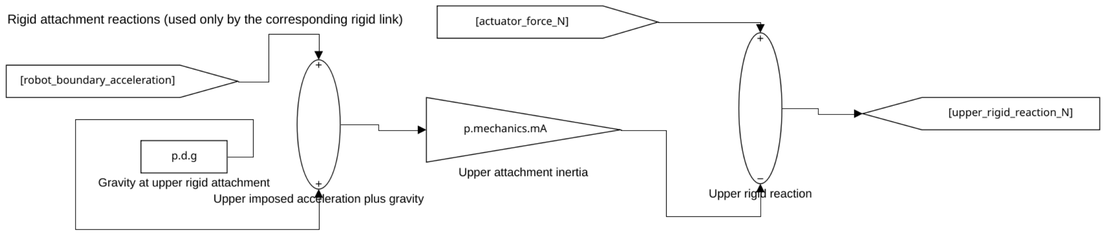


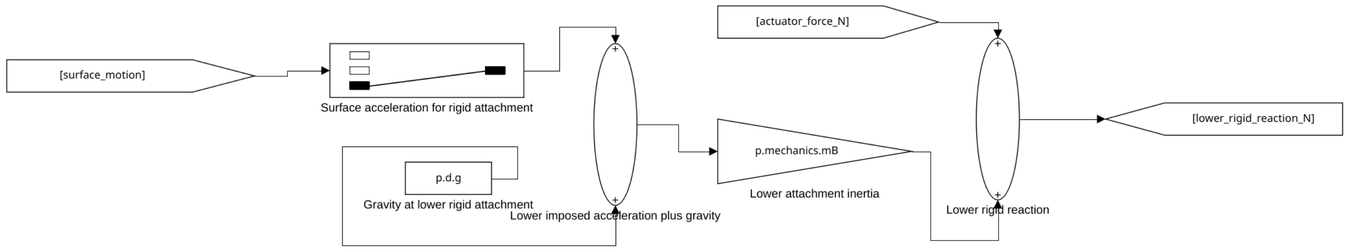






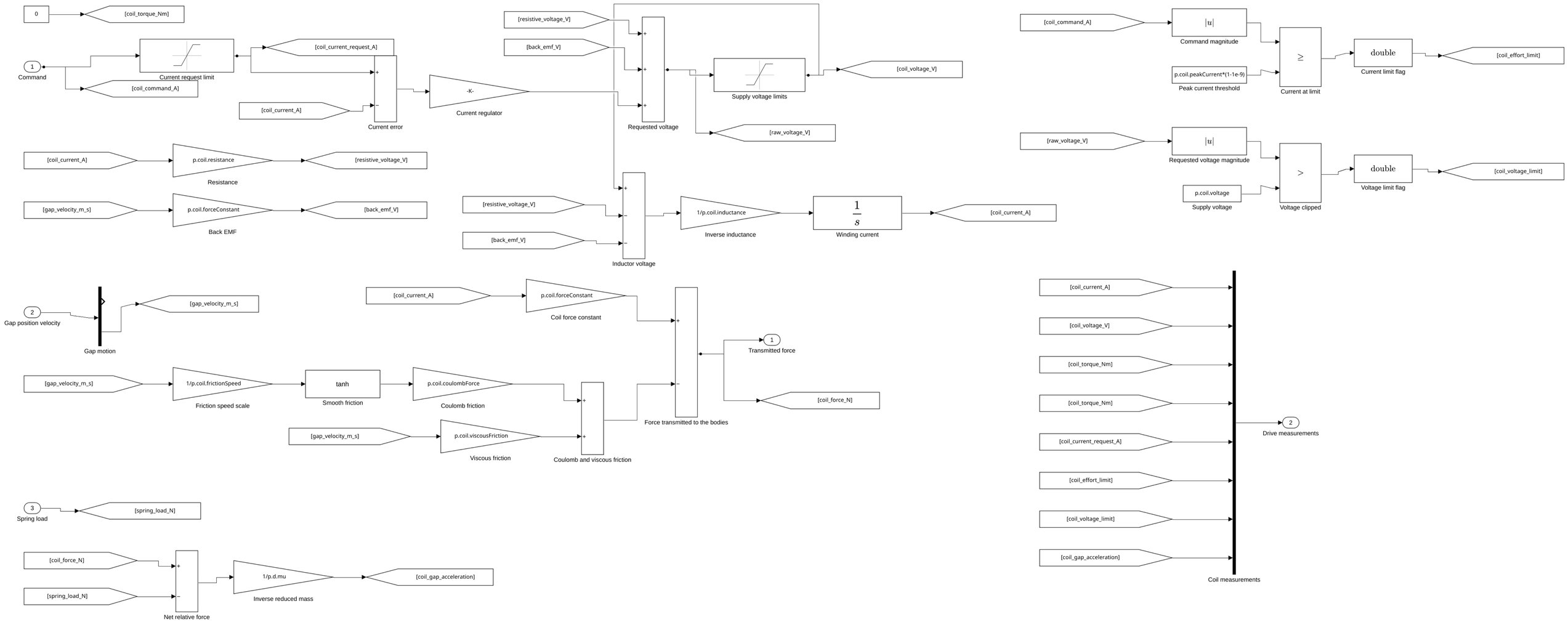




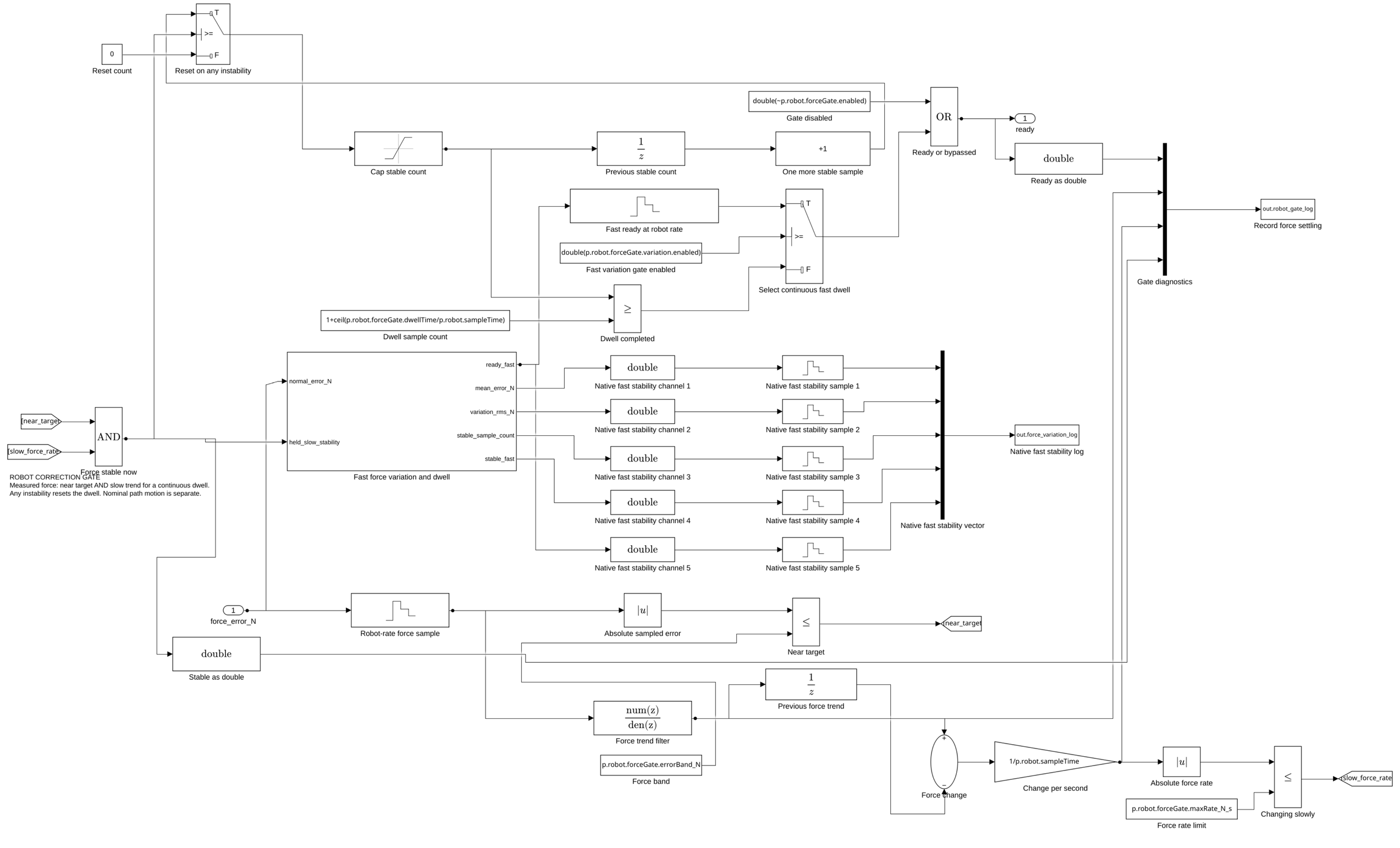


11. acf_force_control_simple/Actuator and RABS mechanics/Actuator/Voice coil

VOICE COIL — current regulator and winding, guide friction, native diagnostics



12. acf_force_control_simple/Recenter travel/Force settling gate



13. acf_force_control_simple/S accelerometer and ADC

